# Title

A Guided GeoAI Research Paper Template (7,000-word target)

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**TEACHING TEMPLATE.**

**Abstract**

**Student instruction:** I like the Nature Summary Paragraph style, so go look at this. Basically write 200 words at the end after completing the main paper. Include as many key statistics as possible, and only the most important, as you are spoonfeeding the reader what to cite. Avoid unsupported significance claims.

**Keywords:** GeoAI; remote sensing; supervised learning; reproducibility

## Word Budget

| Section | Target words |
|---|---|
| 1. Introduction | 900 |
| 2. Literature Review | 1,800 (maximum 2,000) |
| 3. Method | 1,600 |
| 4. Results | 1,200 |
| 5. Discussion | 1,100 |
| 6. Conclusion | 400 |
| **Total** | **7,000** |

**Student instruction:** These are rough planning allocations, not strict sectional limits. In the AI age, fewer words is by far better! And also, make sure you include a high density of both statistics [e.g., in the intro, lit review, results, discussion, conclusion, abstract], and also citations.

If you are stuck at first for words, just bullet point your thinking in each section.

## 1 Introduction

**Student instruction:** Target about 900 words here. Build a narrowing argument from broad significance down to a specific practical or scientific problem. Try to cite what prior work already shows, and then point

to an unresolved gap to justify why this specific analysis is needed. Support claims with citations (as many as possible). The final two paragraphs have specific functions, e.g., (a) state numbered research questions explicitly, and then (b) sketch the organization of the paper. Do not put the roadmap before the RQs.

**Example Research Questions.** This paper addresses the following research questions, including: (RQ1) What overall and class specific prediction performance is obtained from a baseline model using raw spectral bands? (RQ2) By how much, in absolute and relative terms, does adding derived spectral indices change performance? and (RQ3) How does performance vary across defined spatial or sample subsets? Each RQ should be empirically answerable by a result shown later in the paper.

**Example Paper Organization Paragraph.** Section 2 next reviews research on satellite land cover classification and model validation, before Section 3 formalizes the modeling and evaluation approach we intend to use to answer the research questions. Section 4 then reports results organized by research question, before Section 5 interprets these findings against prior literature and considers wider implications. Finally, Section 6 concludes with the contribution and summarizes the principal quantitative results.

# 2 Literature Review

**Student instruction:** Target 1,500 words in the first instance, with a maximum 2,000, but with as many citations as possible. Aim for a dense, relevant and accurate citation base, cite multiple genuine sources per thematic claim where appropriate, but NEVER pad the list with unread or irrelevant references. Organize by ideas and debates, not one paragraph per paper. Close with an explicit synthesis of the gap that motivates your RQs. The citations here are a starting sample, not a comprehensive review.

## 2.1 The substantive problem and evidence base

Define the applied context, so what exactly is being mapped, why it matters, what geographic or temporal scales dominate current research, etc.. Also, make sure you compare what studies agree on and where their findings depend on sensor characteristics or geographical transfer.

## 2.2 Alternative models and theoretical approaches

Explain competing modeling families and justify the one used here.

## 2.3 Feature construction and representation

Review why raw spectral inputs and transformed indices may encode overlapping or complementary information. Define what prior comparisons actually tested in terms of features, label quality, spatial/temporal domain and metrics. Summarize contradictory evidence. Cite the specific empirical papers supporting each comparison rather than reusing a generic method citation.

## 2.4 Synthesis and research gap

Finish with a short synthesis paragraph, including (i) what is established, (ii) what remains contested or untested, (iii) how the proposed model plus empirical study can address the specific gap. This is

the bridge to the methodology, not an additional introduction.

# 3 Method

**Student instruction:** Target 1,600 words. Present the *theoretical mathematical specification first*, then define model variants, training/inference and evaluation, assumptions, and a flow diagram. Place the actual empirical dataset, case study and implementation specific input details in the *last method subsection* (as this is lower value applied information). Never report findings in Method. Up to the end of the method, we should still be talking in future tense, not past tense.

## 3.1 Conceptual and mathematical formulation

[Enter your formal model here.]

## 3.2 Model configurations and hyperparameters

[Discuss the specific model variants and selection procedures, plus hyperparameter choices, etc.]

## 3.3 Training and validation

State train/validation/test partitioning and steps taken to avoid leakage, including spatially separated evaluation if appropriate. Also specify precision, recall, class-specific F1, sample counts and uncertainty intervals as appropriate.

## 3.4 Analytical workflow

Figure 1 is an example schematic. You should think how you can make something like this for your own work (but also better, e.g., introduce nice color, graphics, etc). Distinguish data inputs, model operations, and outputs. We want to show an analytical sequence.

## 3.5 Empirical case study, datasets and parameterization

**Student instruction: This is intentionally the final Method subsection.** Only here identify the study location, sampling frame, spatial/temporal coverage, named datasets, acquisition dates, band definitions, labels, coordinate reference system, resolution, preprocessing, sample counts, missing data procedures, software versions, model parameter values and access/citation links. Include a study area map if appropriate.

# 4 Results

**Student instruction:** Target 1,200 words. Structure subsections by RQ if that is the clearest mapping. Provide multi-panel quantitative graphics with informative captions and consistent scales; refer to panel (a), (b), etc. Explicitly report *overall magnitude*, *absolute difference* (including units or percentage points), and *relative difference* (with reference denominator). Write in dry, neutral terms, which might include little wider explanation, no conjecture or advocacy, no literature comparison or causal claims, etc. These belong in Discussion.

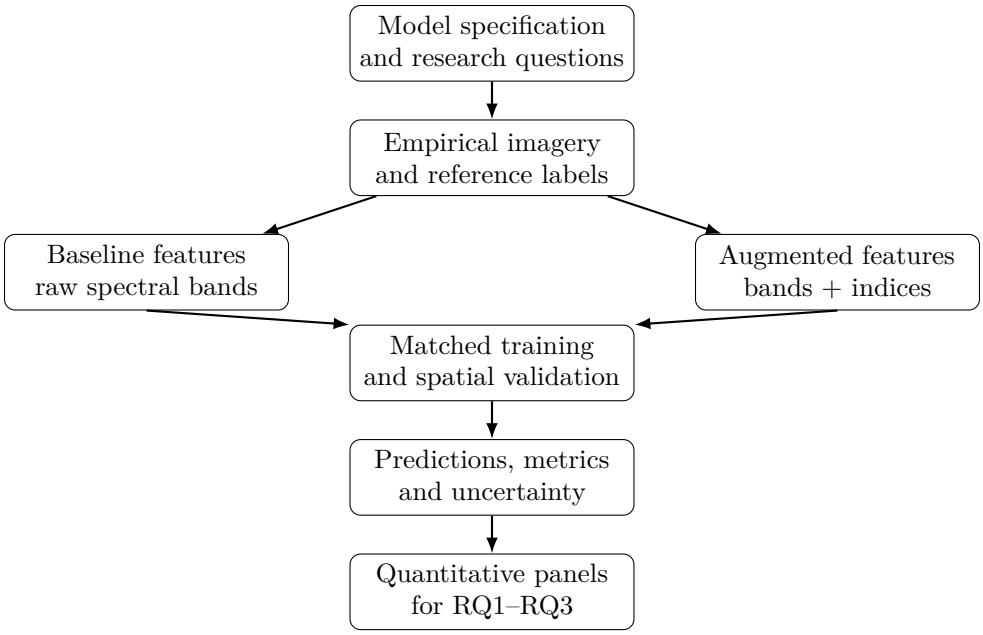



Figure 1: Illustrative workflow. Boxes show theoretical specification, empirical inputs, matched model comparisons and research question quantitative outputs.

### 4.1 RQ1: Baseline performance

A completed paragraph would report test observations, overall accuracy and class performance, each keyed to a figure or table.

You need at least one figure for each RQ, e.g., a panel plot showing overall metrics with a minimum structre of 2x2 plots.

### 4.2 RQ2: Absolute and relative changes

**Fictional illustration only.** "Augmented-model accuracy is set to 0.88 (versus 0.83 baseline), a difference of **5 percentage points** or approximately **6.0% relative to baseline** $(5/83 \times 100)$".

### 4.3 RQ3: Spatial or subgroup variation

For this question, compare specific zones, dates, class frequencies, etc., or other predefined subsets. Give magnitudes and differences. Do not introduce an RQ that the data cannot answer (this goes back to our discssion on falsifiability).

## 5 Discussion

**Student instruction:** Target 1,100 words. Begin with a quick 1-2 sentence intro (In this section the results are reported by...). Then I like to restate the research questions and the purpose of answering them, so we can systematically address each one by interpreting the evidence for each RQ against the literature reviewed in Section 2. Explain whether results support or challenge prior findings, mechanisms that could account

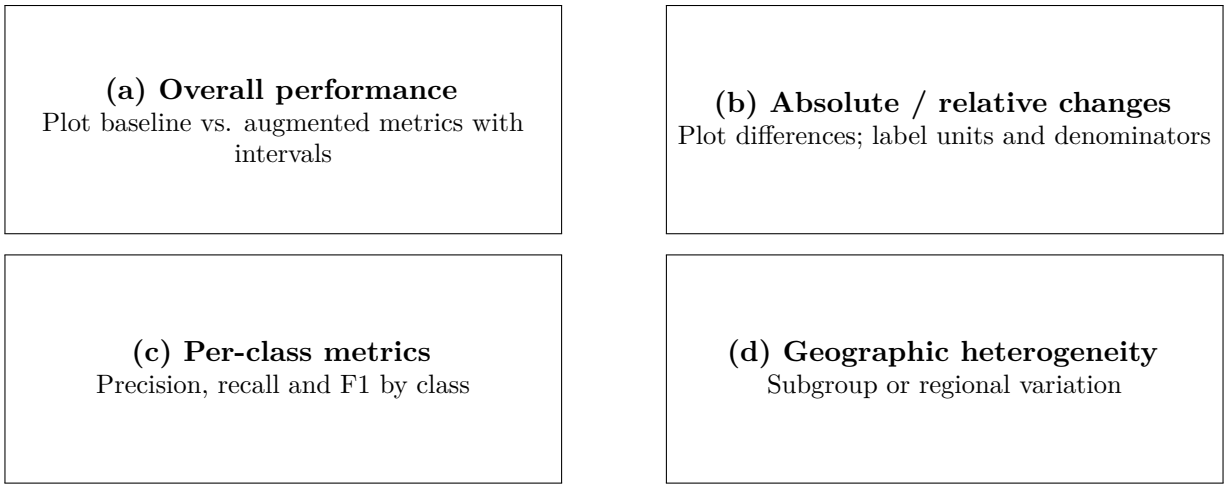



Figure 2: **Four-panel figure planning guide, not generated results.** Replace each labeled box with a quantitative plot derived from actual analysis. Specify panels, samples, metric units, uncertainty, and data sources in the final caption.

for patterns (without overstating causality), implications, generalizability and limitations. Add citations to relevant comparisons. Reserve interpretation for this section rather than Results.

### 5.1 Revisiting the research questions

The study set out to answer three questions, including baseline predictive performance (RQ1), the absolute and relative effect of feature augmentation on measured performance (RQ2), and variation across subsets (RQ3). **Fictional illustration only.** The demonstration values imply only a hypothetical improvement and cannot establish a real effect.

### 5.2 Interpretation against existing evidence

Compare actual patterns with the competing claims in the literature. Explain where your findings add an empirical case, test an existing assumption or produce a methodological contribution. Cite precise and relevant studies, not simply the same review article in every paragraph.

### 5.3 Limitations, implications and next steps

Assess representativeness, sampling choices, measurement error, label quality, domain shift, leakage, uncertainty and model assumptions. Separate limitations that affect internal validity from those affecting generalization. Identify defensible implications based on the measured evidence and practical next steps.

## 6 Conclusion

**Student instruction:** Target 400 words. **Open by stating what the paper did and what it contributes to the literature.** Then give the most important and interesting *quantitative* findings, with

overall values and absolute/relative comparisons where meaningful. Answer the main RQs directly, and then finish with a concise limitation and implication for future research. Do not introduce new literature, methods or results, and avoid vague closing claims.

This paper has illustrated a framework for testing the incremental predictive performance of spectral feature engineering in a supervised GeoAI classification system. Its intended contribution is a transparent comparison of defined feature sets under a common evaluation design. **Fictional illustration only.** In the invented example, accuracy increases from 0.83 to 0.88, a five-percentage-point gain (approximately 6.0% relative to the baseline). A submitted manuscript must substitute verified results, explain the contribution in relation to actual published work, and close with a defensible implication.

## Data and Code Availability

**Student instruction:** Link to real data used, code (e.g., GitHub), reproducible virtual environment, permitted access restrictions, etc.

*This is a fictional example, so see my published work for examples of real data and code.*

## AI Use and Verification

**Student instruction:** Explain any coding or writing assistance, what was checked independently, and how you verified citations, results and figures. Follow course policy. I advise you only use it for idea generation, coding, grammar/typo checking, etc. (so not writing the actual content, or interpretting content, etc.).

## References


Breiman, L. (2001). Random Forests. *Machine Learning*, 45, 5–32.

Belgiu, M. and Drăguț, L. (2016). Random forest in remote sensing: A review of applications and future directions. *ISPRS Journal of Photogrammetry and Remote Sensing*, 114, 24–31.

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Roberts, D. R. et al. (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8), 913–929.

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